Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit. Questions refer to the nonindented paragraph above unless otherwise stated.

Hope student researchers want to determine if students yawning (when being yawned at) is associated with the time of day (morning, afternoon, and evening). The students found the following results from yawning in front of people around campus:

- Morning: 25 of the 39 people yawned back
- Afternoon: 15 of the 38 people yawned back
- Evening: 29 of the 44 people yawned back

1. When this test is run, a p-value of 0.03 is found. What would be an appropriate conclusion?

   (a) Strong evidence exists that the probability a person yawns is different between all three times of day: morning, afternoon, and evening.
   (b) Strong evidence does not exist of an association between time of day and whether a person yawns.
   (c) Strong evidence exists of an association between time of day and whether a person yawns.
   (d) Strong evidence of no association between time of day and whether a person yawns.

   Correct answers: (c)

2. After a chi-square test, follow-up confidence intervals for all pairwise differences in proportions are found as follows:

   - 0.0299 to 0.4627: Morning – Afternoon
   - -0.2237 to 0.1876: Morning – Evening
   - -0.4736 to -0.0551: Afternoon – Evening

   Based off of these intervals, which of the following conclusions are correct?

   (a) The probability of yawning in the morning differs significantly from the probability of yawning the afternoon.
   (b) The probability of yawning in the morning does not differ significantly from the probability of yawning in the afternoon.
   (c) The probability of yawning in the morning differs significantly from the probability of yawning in the evening.
   (d) The probability of yawning in the afternoon does not differ significantly from the probability of yawning in the evening.

   Correct answers: (a)

3. Can telling a joke affect whether or not a customer in a coffee bar leaves a tip for the waiter? A study investigated this question at a coffee bar in France. The waiter randomly assigned coffee-ordering customers into three groups: one received a card telling a joke with their bill, another group received a card containing an advertisement for a local restaurant, and the third received no card. The results are shown below.

	Joke Card	**Ad Card**	**No Card**	**Total**
Left Tip	32 (0.40)	16 (0.20)	24 (0.30)	72
No Tip	48	64	56	168
Total	80	80	80	240

   (a) Yes, since 80 is larger than 20.
   (b) Yes, since 240 is larger than 20.
   (c) No, the sample is not a random sample of customers.
   (d) Yes, since each cell in the table has at least 10 observations.

   Correct answers: (d)

4. Suppose the chi-square statistic for a test was 6.991. If instead, the chi-square statistic had been 5.332, would the p-value be smaller, larger, or remain the same?

   (a) Smaller
   (b) Cannot be determined
   (c) Larger
   (d) Remain the same

   Correct answers: (c)

5. 3) Why do we do overall tests when comparing multiple proportions and not just do the follow-up confidence intervals?

   (a) Doing an overall test allows us to keep the probability of a type I error at 5% or whatever significance level we would like.
   (b) Doing an overall test allows us to quickly get our results, while the follow-up confidence intervals are very time-consuming.
   (c) Doing an overall test allows us to see exactly which group is significantly different from which other groups.
   (d) Doing an overall test will more likely lead to significant results than just doing the follow-up confidence intervals.

   Correct answers: (d)